📘 PRODUCT REQUIREMENTS DOCUMENT (PRD)

VID – Virtual Identification System (School ERP Platform)

1. 📌 Product Overview

VID (Virtual Identification) is a **multi-tenant, role-based School ERP platform** designed to manage multiple educational institutions under a unified, scalable, and secure architecture.

It provides:

- Centralized governance (Super Admin)
- Institution-level autonomy
- Modular workspace-based operations
- High scalability and tenant isolation

2. 🎯 Objectives

- Enable **multi-institution management** from a single platform
- Provide **role-based workspace access control (RBAC)**
- Ensure **data isolation + security (multi-tenant architecture)**
- Deliver **modular, scalable, and configurable ERP system**
- Support **future extensibility (plug-and-play workspaces)**

3. 🏗️ System Architecture

3.1 Multi-Tenant Model

- **Single Database Cluster**
- **Shared Infrastructure**
- **Tenant Isolation via:**
  - institution_id
  - workspace_id

3.2 Data Layer Design

- Row-Level Security (RLS)
- Indexed queries for performance

- Logical segregation per institution
- High availability + optimized queries

4. 🧩 Hierarchical Structure

Super Admin (VID Control Panel)

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Institutions (Tenant Level)

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Institution Admin (Main Admin)

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Workspace Admins (Role-Based)

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End Users (Faculty, Students, Staff)

5. 👑 Super Admin (VID Control Panel)

Responsibilities:

- Manage Institutions
- Subscription & Plans
- Global Settings
- Monitoring & Analytics
- System Health & Logs

Modules:

- Dashboard
- Create Institution
- Global Reports
- System Configuration

📌 From your document:

- Dashboard, Create Institution, Login

6. 🏫 Institution Layer

Each institution acts like an **independent tenant (like an apartment flat)**:

- Custom features per institution
- Independent configuration
- Separate users, roles, and workflows

Institution Admin (Main Admin)

- Full access to all workspaces
- Creates users and assigns roles
- Configures institution settings

7. 🔐 Role-Based Access Control (RBAC)

Access is controlled based on:

- Role (Principal, Accountant, Faculty, etc.)
- Workspace assignment

Examples:

- **Principal** → Academics + Examination
- **Accountant** → Payments + Finance
- **Faculty** → Courses + Timetable + Students

8. 🧠 Request Flow Architecture

User Request

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API Gateway

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Authentication Layer

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Tenant Resolver (institution_id)

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Business Services (Workspace Logic)

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Database Query (Filtered by tenant_id)

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Response

9. 🧱 Core System Modules

From your document:

Core Features:

- Login System
- Notice Board
- Timetable
- Security Layer
- Database Connectivity

📌 Modules include:

- notice_routes
- timetable_service
- authentication logic

10. 🧩 WORKSPACES (Complete System Modules)

This is the **heart of VID** — modular workspace system.

🔹 10.1 Academic Coordinator Workspace

Features:

- Academic Setup
- Courses Management
- Timetable
- Assessments
- Activities
- Principal Dashboard

📌 Includes 13+ modules

🔹 10.2 Admission Officer Workspace

Features:

- Prospects Management
- Applications Tracking
- Admissions Workflow
- Configurations

🔹 10.3 Common Workspace (Student Central)

Acts as Shared Layer:

- Student Central
- Payments
- Examination Access
- HRMS
- User Management System
- Notice Board

🔹 10.4 Course Coordinator Workspace

- Course Allocation
- Course Listing
- Academic Mapping

🔹 10.5 Employee Workspace

- Attendance
- Leaves

- Salary
- Expenses
- Shift Management

🔹 10.6 Examination Workspace

- Results
- Tests
- Reports
- Previous Papers
- Analytics (toppers, averages)

🔹 10.7 Faculty Workspace

- Courses
- Timetable
- Mentees
- Question Banks
- LMS
- Profile Management

🔹 10.8 Hostel Admin Workspace

- Hostel Management
- Attendance
- Gate Pass
- Requests & Reports

🔹 10.9 Payment Administrator Workspace

⚠️ Highly critical financial module

Features:

- Academic Fees
- Online Transactions
- Scholarships
- Concessions
- Reports & Statements

- Fee Cards

📌 Contains 15+ modules

🔹 10.10 Sports Officer Workspace

- Event Management
- Team Handling
- Performance Tracking
- Achievements & Certificates
- Calendar & Reports

🔹 10.11 Student Workspace

Student Dashboard includes:

- Attendance
- Fees & Payments
- Exams
- LMS
- Timetable
- Chatbot
- Profile

🔹 10.12 Team Owner Workspace (Super Admin Equivalent)

- Institution Management
- Organization Settings
- Apps & Integrations
- Data Management

🔹 10.13 Transport Coordinator Workspace

- Bus Management
- Routes
- Boarding Points
- Route Assignments
- Analytics

🔹 10.14 User Management System (Cross-Workspace)

- User Creation
- Role Assignment
- Permission Mapping
- Identity Handling (VID Profiles)

11. ⚙️ Infrastructure Design

Backend:

- API-based architecture (routes.py, services, models)
- Modular service layers per workspace

Frontend:

- Workspace-based UI
- Dashboard-driven navigation
- Role-specific interfaces

12. 🚀 Key Benefits

Scalability

- Supports unlimited institutions

Security

- Tenant isolation (RLS)
- Secure authentication

Performance

- Indexed queries
- Optimized data retrieval

Cost Efficiency

- Shared infrastructure (multi-tenant)

Maintainability

- Modular workspace architecture

Future Ready

- Easy to add new workspaces
- Plug-and-play modules

13. 📈 Future Enhancements

- AI-based analytics dashboards
- Smart attendance (IoT / QR integration)
- Mobile-first application
- Marketplace for add-on modules
- Integration with payment gateways & maps

14. 📄 Summary

VID is not just an ERP — it is a **platform architecture**:

- Multi-tenant ✅
- Role-driven ✅
- Modular ✅
- Scalable ✅
- Secure ✅

It behaves like:
👉 **"ERP-as-a-Service for Educational Institutions"**